P2-L1c-L4 - B1.1-ProbBasics

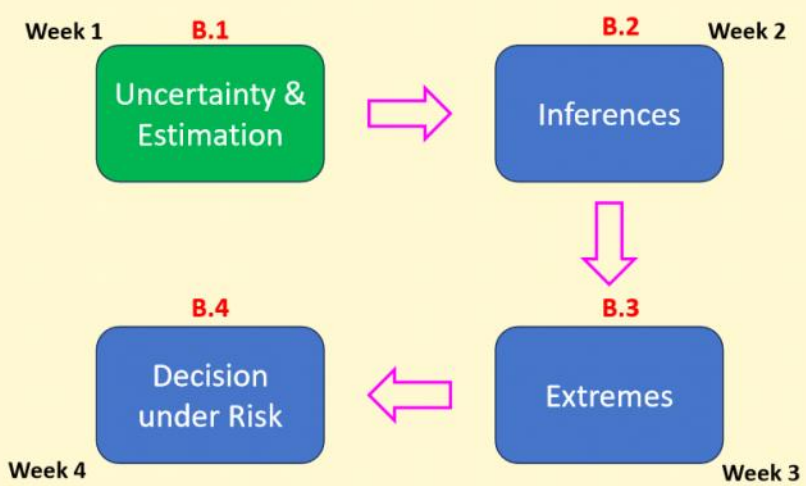
Tuesday, September 1, 2026 10:17 AM

CV 5101
Modeling, Uncertainty, and Data for Engineers

(July – Nov 2026)

Dr. Prakash S Badal

Module Overview



A crash course in probability, probabilistic models, and probabilistic methods.

Check-in with teach-book

https://mude.citg.tudelft.nl/book/2024/

- Q1 Topics (Chapters 5, 6, 2, and 3)
  - Q1C5 Univariate continuous distribution
    - Q1C5.1 PDF/CDF
    - Q1C5.2 Empirical Distributions
    - Q1C5.3 Parametric Distributions
    - Q1C5.4 Fitting a Distribution
  - Q1C6 Multivariate Distributions (briefly)
  - Q1C2 Propagation of Uncertainty
  - Q1C3 Observation Theory: least-sq., Hyp. Test, Conf. Intervals
- Q2 Topics (Chapter 7 and 8)
  - Q2C7 Extreme value theory: GEV, return period, POT
  - Q2C8 Risk and decision making (CBA)
- Fundamental Concepts
  - Chapter 6, 7, 8, 9. Probability basics, rv, z- and t-tables
- Programming
  - Fundamental Concepts → Chapter 10

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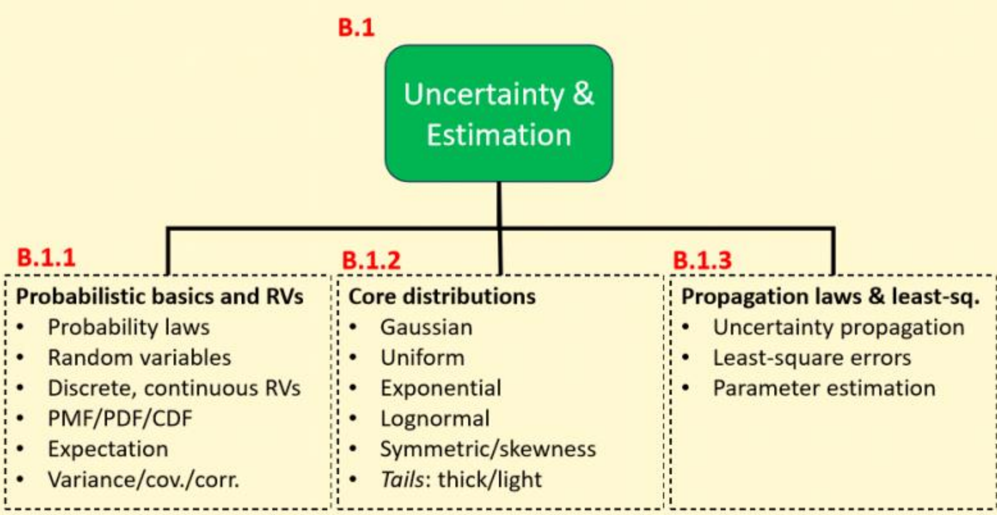
Module Overview



B.1.1 Probability basics & random variables

Birthdays!

Birthday-Birthday

Jan 4 14	**Feb** 22 23	**Mar** 12 14
Apr 6 15 17	**May** 6 11 12 14 23	**Jun** ~~26~~ ~~26~~
Jul	**Aug**	**Sep**
Oct	**Nov**	**Dec**

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Question 1

Probability that at least two people in this room share the birthday?

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#cv5101

Birthday Problem

Pr(Two people share birthdays) = $1/365$

Pr(2 people share birthdays in a class of 40)

$$= 1 - \frac{364 \times 363 \times \ldots \times (365 - n + 1)}{365 \times 365 \times \ldots \text{ 40 times}}$$

for $n = 50$, Pr(at least 2 shared birthdays) ≈ 0.97

$n = 23$, Pr(at least 2 shared birthdays) ≈ 0.5

Birthday attack $\leftarrow$ collision attack

$2^{64} \longrightarrow 2^{32}$

Intuition can be misleading.

Settlers of Catan

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Practice question

- Work with the another person around you.
- You throw 2 dice;
- add the values of each die
- List all the possible outcomes

Q1. What is the probability that the sum of the two dice will be 6?

Q2. What is the probability that you get at least one 8 in a single round consisting of four players? $1-(31/36)^4$

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Probability axioms

S: Sample Space

$$Pr(S) = 1$$

E: An Event

$$0 \leqslant Pr(E) \leqslant 1$$

mutually exclusive

E_1, E_2: Two events s.t. $E_1 \cap E_2 = \emptyset$

$$Pr(E_1 \cup E_2) = Pr(E_1) + Pr(E_2)$$

Probability rules

$$Pr(E_1 \cup E_2) = Pr(E_1) + Pr(E_2) - Pr(E_1 \cap E_2)$$

Union

$$Pr(\bar{E}_1) = 1 - Pr(E_1)$$

Complement

$$Pr(E_1 \cup E_2 \cup E_3) = Pr(E_1) + Pr(E_2) + Pr(E_3) - Pr(E_1 \cap E_2) - Pr(E_2 \cap E_3) - Pr(E_3 \cap E_1) + Pr(E_1 \cap E_2 \cap E_3)$$

Inclusion-Exclusion principles

$\equiv Pr(E_i \cap E_j)$

$$Pr\left(\bigcup_{i=1}^{n} E_i\right) = \sum Pr(E_i) - \sum_{i<j} Pr(E_i E_j) + \sum_{i<j<k} Pr(E_i E_j E_k) - \dots + (-1)^{n-1} Pr(E_1 E_2 \cdots E_n)$$

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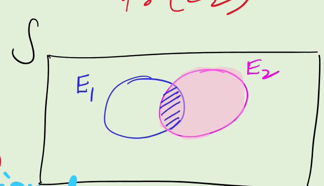
Probability rules: conditional prob.

- Conditional probability: $Pr(E_1 \mid E_2) = \dfrac{Pr(E_1 \cap E_2)}{Pr(E_2)}$

Probability of E_1 given that E_2 has occured



- Multiplication rule:

$$Pr(E_1 \cap E_2) = Pr(E_1 \mid E_2) \cdot Pr(E_2)$$

conditional

$$Pr(E_1 \cap E_2 \cap \cdots \cap E_n) = Pr(E_1 \mid E_2 E_3 \cdots E_n) \cdot Pr(E_2 \mid E_3 E_4 \cdots E_n) \cdot \cdots Pr(E_{n-1} \mid E_n) \cdot Pr(E_n)$$

Marginal

- Two events are *called* statistically independent (SI), if

$$Pr(E_1 \mid E_2) = Pr(E_1)$$

$$\Rightarrow \frac{Pr(E_1 \cap E_2)}{Pr(E_2)} = Pr(E_1) \Rightarrow Pr(E_1 \cap E_2) = Pr(E_1)\, Pr(E_2)$$

Non-zero unless either $Pr(E_1)$, $Pr(E_2)$ is zero

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L2

Sep 2, 2026

Messages from the Last Class

- **Message 1:** *Space Shuttle Challenger Disaster*

If everything is important, then nothing is!

- **Message 2:** *Basic Probability*

→ you need to revise prob & stats

→ Sample space, S & Event, E

$Pr(S) = 1$ $\quad 0 \leq Pr(E) \leq 1$

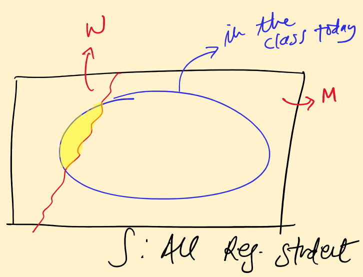
If E_1 & E_2 are exclusive $E_1 \cap E_2 = \emptyset$

$Pr(E_1 \cup E_2) = Pr(E_1) + Pr(E_2)$

- **Message 3:** *Conditional prob.*

→ $Pr(A|B) = \dfrac{Pr(A \cap B)}{Pr(B)}$

→ Conditioning is equivalent to Reducing the sample space.



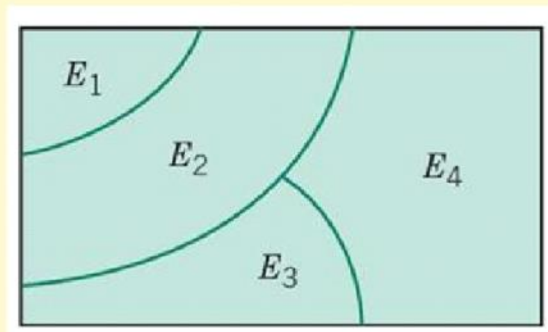
Set Theory: Mutually Exclusive Events

If $E_1, E_2, \ldots, E_n$ are such that

$$E_i \cap E_j = \emptyset$$

then

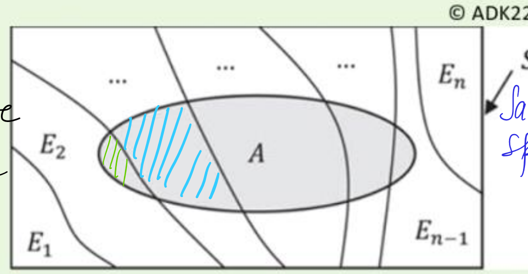
$$Pr(E_1 \cup E_2 \cup \cdots \cup E_n) = Pr(E_1) + Pr(E_2) + \cdots + Pr(E_n)$$



Probability rules: total probability rule

- If $E_1, E_2, \ldots, E_n$ are n MECE events, Mutually exclusive & collectively exhaustive

$E_1 \cup E_2 \cup \cdots \cup E_n = S$



Sample space

$$Pr(A) = Pr(AE_1) + Pr(AE_2) + \cdots + Pr(AE_n)$$

$$= \sum_{i=1}^{n} Pr(AE_i)$$

$$= \sum_{i=1}^{n} Pr(A|E_i) \cdot Pr(E_i)$$

Total Probability Rule

(At home) Probability rules: total probability rule

The probability that a building will sustain damage due to an earthquake depends on the intensity of shaking. Suppose the intensity is discretized into the states of weak, moderate, and strong. Assume it is determined that the probability of damage to the building is 0.01, 0.10, and 0.60 for the three levels of shaking intensity, respectively. Furthermore, based on the history of past earthquakes in the region, it is estimated that, for a randomly occurring earthquake, the probabilities for the three levels of shaking intensity at the site of the building are 0.90, 0.08, and 0.02, respectively. We are interested in the probability that the building will sustain damage during the next earthquake in the region.

Pr(D|W)

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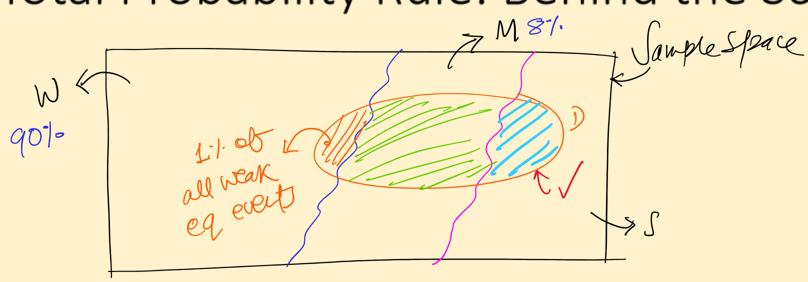
Total Probability Rule: Behind the Scenes!



Pr(Shaking) — W 90%, M 8%, S 2%

Seismologists

Pr(Damage | Shaking) — W 1%, M 10%, S 60%

Structural Engineers

2%   0.9%   0.8%

90% × 1% + 8% × 10%

Pr(W) Pr(D|W)   Pr(M) Pr(D|M)

\+ 1.2%

2% × 60%

Pr(S) Pr(D|S)

= 2.9%

Total probability rule

→ Breaking down the calculation Pr(A) into computing the conditional Pr(A | Ei)

→ Because conditionals are often easy to calculate

→ Clever partitioning of sample space

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Partitioning Complex Structural Systems

probability that Chennai Airport is closed tomorrow?

Weather prediction
- → No rain
- → Moderate rain
- → Heavy rain
- → Cyclone

Prob. that a high-rise building cannot be evacuated safely during a fire?

Fire initiation
- → Basement
- → Lower floor
- → Middle floor
- → upper floor

Probability rules: Bayes' rule

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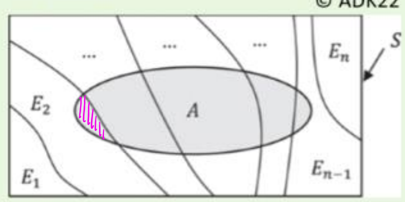
- If $E_1, E_2, \ldots, E_n$ are n MECE events,



$Pr(E_i | A)$ ← Probability of Interest

$$Pr(AE_i) = Pr(E_i | A) \cdot Pr(A) = Pr(A | E_i) \cdot Pr(E_i)$$

$$Pr(E_i | A) = \frac{Pr(A | E_i) \cdot Pr(E_i)}{Pr(A)}$$

$$= \frac{Pr(A | E_i) \cdot Pr(E_i)}{\sum_{j=1}^{n} Pr(A | E_j) \cdot Pr(E_j)}$$

(At home) Probability rules: Bayes' rule

- Assume in the previous example that the building has sustained damage. What can we learn about the intensity of the earthquake?

$Pr(W|D) = 0.9/2.9 = 31\%$

$Pr(M|D) = 0.8/2.9 = 27\%$

$Pr(S|D) = 1.2/2.9 = 41\%$

Posterior

90% ← $Pr(W)$

8% ← $Pr(M)$

2% ← $Pr(S)$

Prior

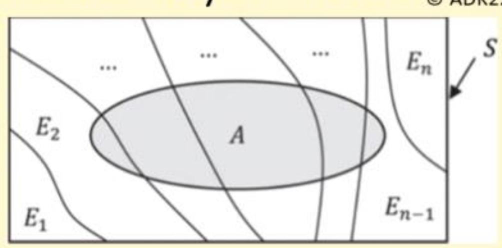
$Pr(W|D)$, $(M|D)$, $(S|D)$

$Pr(W)$, (M), (S)

Posterior = likelihood × Prior

Total Probability Rules & Bayes' rule

$E_1, E_2, \ldots, E_n$ are (MECE) partitions of S.



L3
Sep 3, 2026

Total Probability and Bayes' Rule irl

Total Probability Rule Q) What will happen?

→ Pr (cancer) → Insurance

→ Pr (flight cancelled) → Flight operators

→ Pr (Building collapse) → Design standard

Bayes' Rule Q) Now that "this" has happened, what caused this event?

→ Given that a patient has some symptom

→ Request more tests

→ Treatment/Medication depends on these probs.

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Murder trial of O. J. Simpson

O. J. Simpson

57 languages

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From Wikipedia, the free encyclopedia

"The Juice" redirects here. For other uses, see Juice (disambiguation).

Orenthal James Simpson (July 9, 1947 – April 10, 2024), nicknamed **"the Juice"**,[b] was an American football player, actor, and media personality. He played in the National Football League (NFL) for 11 seasons—nine with the Buffalo Bills—and is regarded as one of the greatest running backs of all time. His success is widely considered to be overshadowed by his two criminal charges for the 1994 murders of his ex-wife Nicole Brown and her friend Ron Goldman, and the contentious criminal trial in which he was acquitted on both counts.

Simpson played college football for the USC Trojans, winning the 1968 Heisman Trophy as a senior. He was selected first overall by the Bills in the 1969 NFL/AFL draft. With the Bills, he received five consecutive Pro Bowl and first-team All-Pro selections from 1972 to 1976. He led the league in rushing yards four times, in rushing touchdowns twice, and in points scored in 1975. Simpson became the first NFL player to rush for more than 2,000 yards in a season—earning him 1973's NFL Most Valuable Player (MVP) award—and he is the only player to do so in a 14-game regular season. He also holds the record for the single-season yards-per-game average, at 143.1.

O. J. Simpson

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Murder trial of O. J. Simpson

Which probability should the jury care about?

Pr(DNA match | innocent)

Or

Pr(guilty | DNA match)

15-20x more

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Murder trial of O. J. Simpson

DNA expert say, that

$$Pr(\text{DNA match} \mid \text{unrelated person}) = \frac{1}{20 \text{ million}}$$

Jury wants to know

$$Pr(\text{guilty} \mid \text{DNA match})$$

Bayes' $Pr(\text{guilty} \mid \text{DNA match}) = \dfrac{Pr(\text{DNA match} \mid \text{guilty}) \cdot Pr(\text{guilty})}{Pr(\text{DNA match})}$

Prosecutor's Bias $Pr(G \mid DNA) \propto Pr(DNA \mid G)$ ← NOT wrong, But incomplete

$Pr(G \mid DNA) \propto Pr(DNA \mid G) \cdot \underline{Pr(G)}$ ← Baseline rate

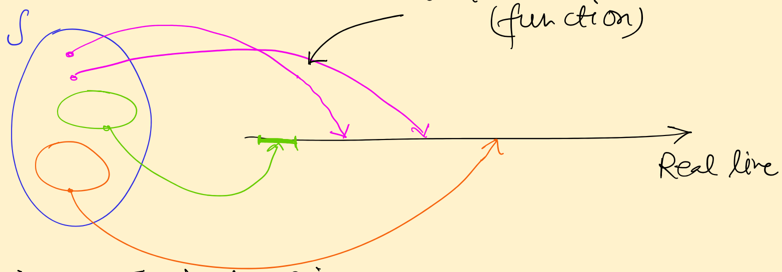
DNA match $\not\Longrightarrow$ Create guilt

100,000,000

	MATCH	NO MATCH
GUILTY	1	~~0~~
INNOCENT	5	~~~100,000,000~~

Random Variables

Random variable



Experiment: Tossing two coins

Sample Space $\{(H,T), (H,H), (T,H), (T,T)\}$

X: Number of heads in two tosses of a coin

$$X(\omega) = \begin{cases} 0 ; & \omega = (T,T) \\ 1 ; & \omega = (H,T) \text{ or } (T,H) \\ 2 ; & \omega = (H,H) \end{cases}$$

↑ Random Variable

RV: misnomer. Neither Random / Variable. It is a function.

Random variable

- Experiment: rolling two 4-faced dice

$X =$ maximum roll

Domain of X? → S

Range of X? → $\{1,2,3,4\}$

$S =$?

$X(w) =$?

$$X(\omega) = \begin{cases} 1, & \omega = (1,1) \\ 2, & \omega = (1,2), (2,1), (2,2) \\ 3, & \omega = (1,3), (2,3), (3,1), (3,2), (3,3) \\ 4, & \omega = (1,4), (2,4), (3,4), (4,1), (4,2), (4,3), (4,4) \end{cases}$$

$Pr(X=1) = 1/16$

$Pr(X=2) = 3/16$

$Pr(X=3) = 5/16$

$Pr(X=4) = 7/16$

$$\sum_{i=1}^{n} Pr(X=\omega_i) = 1$$

Types of RV

Discrete

- Range is finite or countable
  - Grade in this course
  - # of sand particles on Marina
  - Natural numbers

Continuous

- Range is continuous over some interval
  - temperature
  - height
  - weight
  - $x \in (0, 1)$

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L4

Sep 8, 2026

Random variable (rv): PMF, PDF

- For a discrete rv:

Probability mass function (PMF) $P_X(x) = Pr(X = x)$

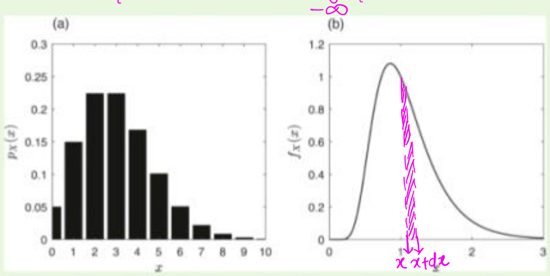
PMF must satisfy: $0 \leq P_X(x) \leq 1$, $\sum_x P_X(x) = 1$

- For a continuous rv:

Probability density function (PDF) $f_X(x)$

$Pr(x < X \leq x+dx) = \int_x^{x+dx} f_X(x)\,dx$

PDF must satisfy: $f_X(x) \geq 0$, $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$



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SIDE NOTE:
for real numbers,
Never compare their equality
~~if x == 5~~
if abs(x-5) << tol ✓

(At home) Random variable (rv): PMF, PDF

- Let X = concrete strength, in the range $[0, 20\ \mathrm{MPa}]$
- Assume a PDF given as:

$$f_X(x) = 0.05 \text{ for } 0 \leq x \leq 20$$

- What is the probability that the strength of a tested concrete is less than 10 MPa?

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Random variable (rv): CDF

- *Cumulative distribution function* (CDF) is defined as:

$$F_X(x) = Pr(X \leq x)$$

- The CDF of a discrete RV X is

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i)$$

$Pr(X \leq x)$

- The CDF of a continuous RV X is

$$F_X(x) = \int_{-\infty}^{x} f_X(x)\, dx$$

$$\rightarrow f_X(x) = \frac{d}{dx} F_X(x)$$

$\rightarrow$ Non-decreasing

$$CCDF = Pr(X > x) = 1 - F_X(x)$$

↳ complementary

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Random variable (rv): CDF

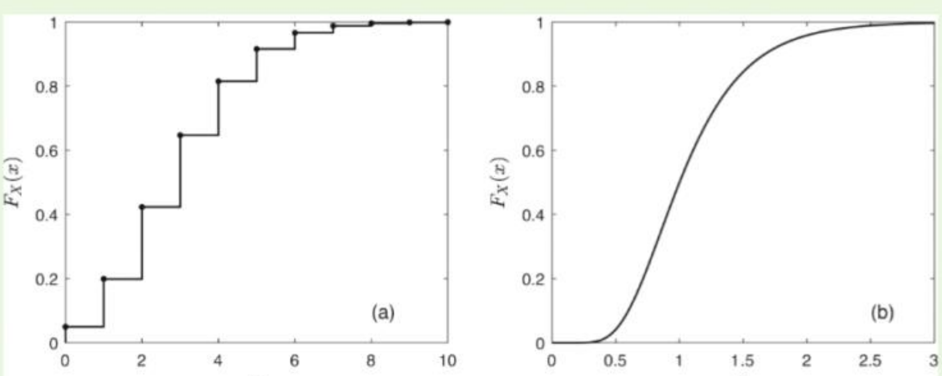
CDF must satisfy: $F_X(-\infty) = 0 \; ; \; F_X(\infty) = 1$

if $x < y \Rightarrow F_X(x) \leq F_X(y)$

	fair	loaded/biased
1	1/6	$\alpha = 1/12$
2	1/6	$3\alpha = 3/12$
3	⋮	α
4	⋮	3α
5	⋮	α
6	1/6	3α

$12\alpha = 1$

$\alpha = 1/12$



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Expectation of RV

Expectation

$$E[X] = \sum_{x_i} x_i \, p_X(x_i) = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

- Let $g(X)$: a function of rv X, with PDF $f_X(x)$.
  Expectation of $g(X)$,

$$\mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

$$\mathrm{E}[\textstyle\sum_i g_i(\mathbf{X})] = \sum_i E[g_i(X)]$$

→ expectation is a linear operator

measure of dispersion

μ

E[X]

measure of centrality

→ deviation $X-\mu$ → $E[X-\mu]=0$

→ Mean Absolution Deviation (MAD)

$E[\,|X-\mu|\,]$

→ Square of deviation → $E[(X-\mu)^2] = \sigma^2$ → Variance

- Expectation/mean of X

$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

Variance

- Variance is the expected value of $(X - \mu_X)^2$,

$$
\begin{aligned}
\sigma^2 &= E[(X-\mu)^2] \\
&= E[X^2 + \mu^2 - 2X\mu] \\
&= E[X^2] + E[\mu^2] - E[2X\mu] \\
&= E[X^2] + \mu^2 - 2\mu^2 \\
&= E[X^2] - \mu^2 \\
&= E[X^2] - (E[X])^2
\end{aligned}
$$

Covariance

X_1 : mean μ_1 ; variance σ_1^2

X_2 : mean μ_2 ; variance σ_2^2

$$
\begin{aligned}
COV[X_1, X_2] &= E[(X_1-\mu_1)(X_2-\mu_2)] \\
&= E[X_1 X_2] - E[X_1]\,E[X_2] \\
&= E[X_1 X_2] - \mu_1\mu_2
\end{aligned}
$$

Correlation

- Correlation coefficient

$$\rho_{12} = \frac{COV[X_1, X_2]}{\sigma_{X_1}\,\sigma_{X_2}}$$

$$COV[X_1, X_2] = \rho_{12}\,\sigma_{X_1}\,\sigma_{X_2}$$

$$-1 \leqslant \rho_{12} \leqslant 1$$

Correlation coefficient is a metric of predictability of expense of one variable, given the expense of another variable.

1 stdev taller than average

Height	Weight	
+1	→ +1	$\rho \to 1$ perfectly +vely correlated
−0.5	→ −0.5	
−2.0	→ −2.0	
+1	→ −1	$\rho \to -1$
−0.5	→ +0.5	
−2.0	→ +2.0	
+1 (one-person × 1000)	→ +1 (200), +2 (200), ±0 (200), −1 (200), −2 (200)	$\rho \to 0$
−0.5	→ +0.5, −0.5	
−2.0	→ +2.0, −2.0	

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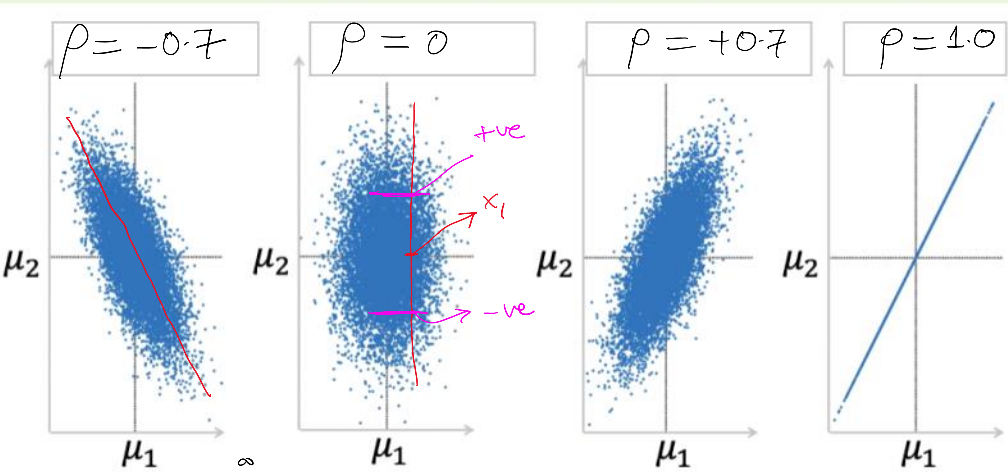
Correlation



Moment $E[g(x)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx$

$g(x) = x^n \Rightarrow E[X^n] = M_n \leftarrow n^{th}$ moment

$g(x) = (x-\mu)^n \Rightarrow E[(X-\mu)^n] = \eta_n \leftarrow n^{th}$ central moment

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Moment

1^{st} moment $\Rightarrow E[X] = \mu$ mean

2^{nd} moment $\Rightarrow E[X^2]$

Central Moments

1^{st} ¢ moment $\Rightarrow E[(X-\mu)] = 0$

2^{nd} ¢ moment $\Rightarrow E[(X-\mu)^2] = \sigma^2$

3^{rd} ¢ moment $\Rightarrow E[(X-\mu)^3]/\sigma^3 = \theta$ skewness

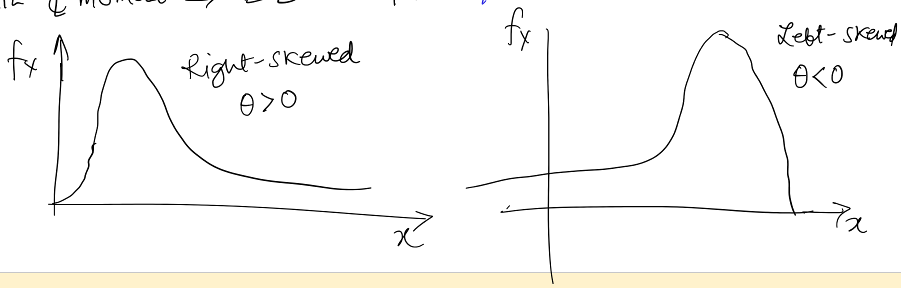
4^{th} ¢ moment $\Rightarrow E[(X-\mu)^4]/\sigma^4 = \kappa$ Kurtosis ← tailedness

3rd ¢ moment $\Rightarrow E[(X-\mu)^3]/\sigma^3$

4th ¢ moment $\Rightarrow E[(X-\mu)^4]/\sigma^4 = \kappa$

Kurtosis ← tailedness

$\kappa > 3$	$\kappa = 3$	$\kappa < 3$
Heavy tail	Normal tail	Light tail



Analogy with Engineering Mechanics

Param.	Mechanics	Probability
Mass	Mass (or area)	Probability
Value	Position x	RV X
Density	Mass density $\rho(x)$	Probability Density Function $f_X(x)$
First Moment	$M_1 = \int x\, dm$ Center of mass, $\bar{x} = \frac{1}{m} \int x\, dm$ Where is the average material located?	Expected value, $\mu = E[X]$ $= \sum x_i\, p_X(x_i)$ $= \int x f_X(x)\, dx$
Second moment	$M_2 = \int x^2\, dm$ Moment of Inertia How widely is material distributed?	Variance, $\sigma^2 = Var(X)$ $= E[(X-\mu)^2]$ $= E[X^2] - \mu^2$ $= \sum (x_i-\mu)^2 p_X(x_i)$ $= \int (x-\mu)^2 f_X(x)\, dx$

Questions, comments, or concerns?